Exercise Session: Conformal Mappings

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In this session, we will explore how to construct conformal mappings between various domains, starting with fractional linear transformations and moving to transcendental functions.

Exercise 1: Warm-up with the Lower Half-Plane

In the lectures, we derived the general formula mapping the Upper Half-Plane ($\mathrm{Im}(z) > 0$) to the Unit Disk ($\mathbb{D} : |w| < 1$). Now, construct a Möbius transformation $w = T(z)$ that maps the **Lower Half-Plane** ($\mathbb{H}^- : \mathrm{Im}(z) < 0$) onto the interior of the Unit Disk.

(a) Choose a specific point z_0 in the lower half-plane to map to the origin $w = 0$.

(b) Build the related Moebius transformation (check your notes!)

(c) Verify your transformation by picking a test point in the lower half-plane and ensuring its image has a modulus strictly less than 1.

Exercise 2: $w = e^z$

Let's analyze the mapping properties of the exponential function $w = f(z) = e^z$.

(a) Write $z = x + iy$ and express w in polar form ($w = \rho e^{i\phi}$). How do x and y relate to ρ and ϕ?

(b) Find the image of the infinite horizontal strip defined by $0 < y < \pi$ and $-\infty < x < \infty$. Sketch the domain and its image.

(c) What is the image of a vertical line segment $x = c$ (where c is a constant) restricted to the strip $0 < y < \pi$?

Exercise 3: $w = \sin(z)$

Let's analyze the mapping $w = \sin(z)$. In the lectures, we mapped a strip of width π. Let's look at a narrower domain. Consider the semi-infinite vertical strip defined by $0 < x < \pi/2$ and $y > 0$.

(a) Expand $w = \sin(x + iy)$ into its real and imaginary parts $u(x, y)$ and $v(x, y)$.

(b) Find the image of the boundary $x = 0, y > 0$.

(c) Find the image of the boundary $x = \pi/2, y > 0$.

(d) Find the image of the boundary segment $0 < x < \pi/2, y = 0$.

(e) Conclude what region in the w-plane represents the image of the interior of this semi-infinite strip. Sketch both domains.

Exercise 4: Composing Mappings

By composing two mappings we have analyzed today, construct a single analytic function $W = F(z)$ that maps the horizontal strip $0 < \mathrm{Im}(z) < \pi$ **onto the interior of the unit disk** $|W| < 1$.